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clustering

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detail

- Overview of Cluster Analysis
- Hierarchical cluster analysis
- Center-based cluster analysis
- Cluster evaluation scale
- k-means clustering practice

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Overview of Cluster Analysis

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The meaning of cluster analysis

• Cluster analysis

– The process of dividing observed objects into several groups based on similarity (affinity) – Individuals

belonging to the same group have high similarity to each other, and differ from individuals belonging to other group

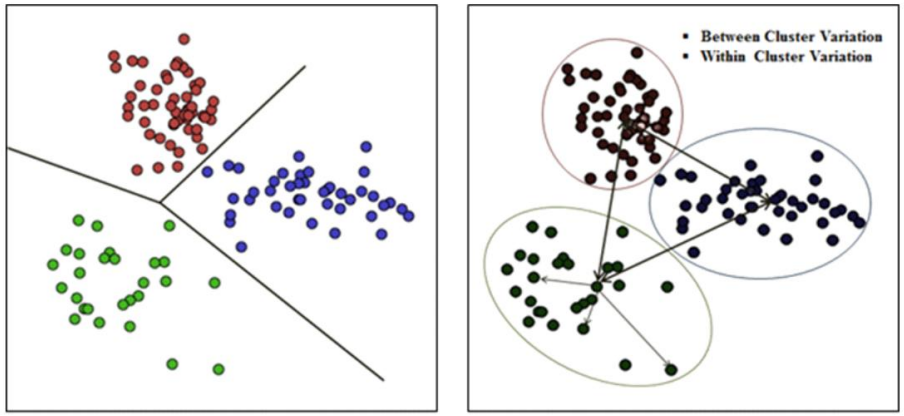
The likelihood of error

must be small. • Difference between

classification and cluster analysis – Classification:

A supervised learning method – Cluster analysis: Forms groups or clusters based on similarity between entities

without prior information about categories; an unsupervised learning method



(a) 분류 분석 (b) 군집 분석

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Purpose of cluster analysis

(1) Search and summary of materials

– Understanding the structure inherent in data as a means of exploratory research

– The task of dividing unmarked objects or instances into homogeneous groups called clusters

– Identifying the characteristics of each cluster helps in understanding and interpreting the structure of the entire data

(2) Used as a pretreatment method

– Use the results of cluster analysis as follows

ÿ Used for automobile insurance claim patterns, land use patterns, and identifying earthquake epicenters.

ÿ Establishing marketing strategies based on market segmentation: For example, a telecommunications company's customer churn model

Establishing prediction models for each cluster during development

ÿ Data Reduction: By remembering only the data of entities or centroids that represent the cluster, to remember

It is widely used for compressing or processing large-scale image data by reducing the amount of data.

ÿ Outlier Detection: Used for Outlier Detection, which does not belong to any cluster

Judging individuals as outliers

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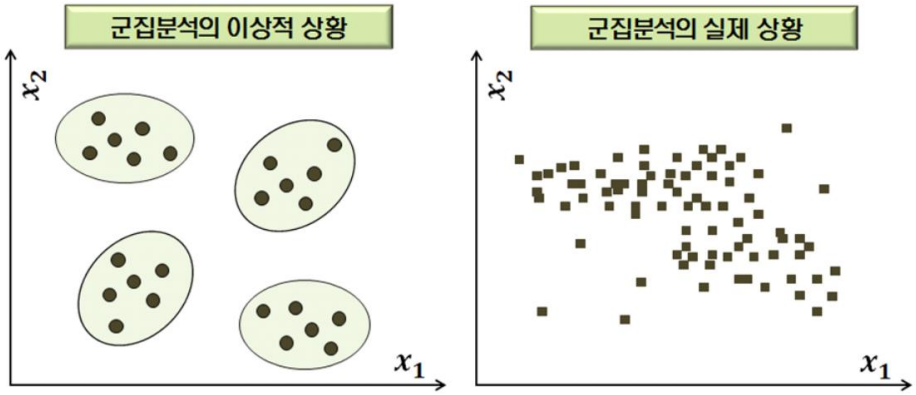
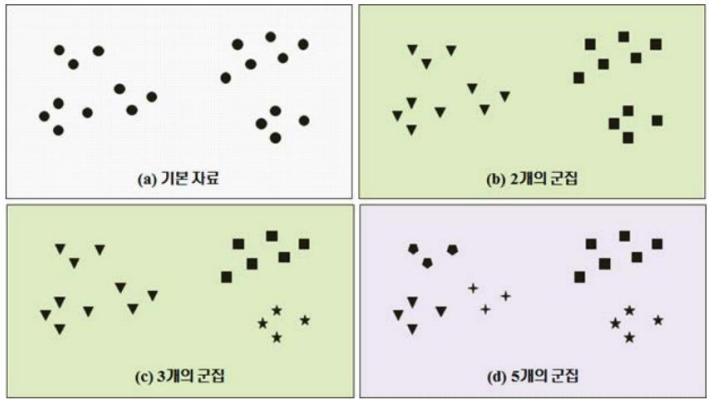
Ambiguity and Subjectivity in Cluster Analysis

- When performing cluster analysis on data where group distinctions are highly ambiguous : – Within-cluster or between-cluster similarity cannot serve as an absolute criterion for judgment. – It is difficult to determine whether the generated clusters are actually inherent in the data or if they are good clusters.



- The evaluation of cluster analysis results is highly subjective
  - It is difficult to rely solely on within-cluster or between-cluster similarity. – A good cluster is one that contributes to achieving the analysis objective.
    It is appropriate to judge based on whether it is a result
  - the meaning of a good cluster changes depending on the nature of the analysis data and what the desired outcome is.



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Procedure for Performing Cluster Analysis

- Data and Clusters

$$D = \begin{array}{c|ccccc} & X_1 & X_2 & \cdots & \cdots & X_m \\ \hline x_1 & x_{11} & x_{12} & \cdots & \cdots & x_{1m} \\ x_2 & x_{21} & x_{22} & \cdots & \cdots & x_{2m} \\ \vdots & \vdots & \vdots & \ddots & & \vdots \\ \vdots & \vdots & \vdots & & \ddots & \vdots \\ x_n & x_{n1} & x_{n2} & \cdots & \cdots & x_{nm} \end{array}$$

$C = \{C_1, C_2, \cdots, C_K\}$

여기서

1) $C_i \cap C_j = \varnothing,\ 1 \le i \ne j \le K$

2) $\bigcup_{i=1}^{K} C_i = D$

- Distance (Similarity)

Measures – Euclidean distance, correlation coefficient, cosine, Mahalanobis distance, etc.

– Standardization is necessary when the unit difference between variables is large *

Distance matrix

$$d_{ij} = d(x_i, x_j) = \| x_i - x_j \|_2 \qquad DM = \begin{bmatrix} 0 & d_{12} & d_{13} & \cdots & d_{1n} \\ d_{21} & 0 & d_{23} & \cdots & d_{2n} \\ d_{31} & d_{32} & 0 & \cdots & d_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ d_{n1} & d_{n2} & d_{n3} & \cdots & 0 \end{bmatrix}$$

- Various clustering algorithms

  – K-Means etc.

- Determining the Number of

  Clusters – The optimal number of clusters cannot be known in advance from the given data. –

  Requires theoretical concepts as well as empirical and practical considerations.

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Hierarchical cluster analysis

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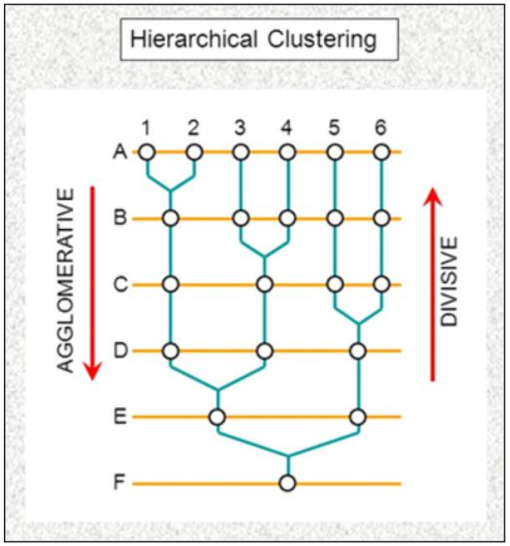
Two approaches to hierarchical clustering analysis

ÿ Agglomerative method: Primarily used for hierarchical clustering analysis – Bottom-up approach

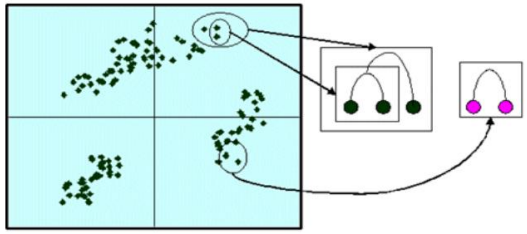
– Starts by treating each object in the data as a separate cluster – Merges the two closest clusters at each step – Continues until all objects form a single cluster ÿ

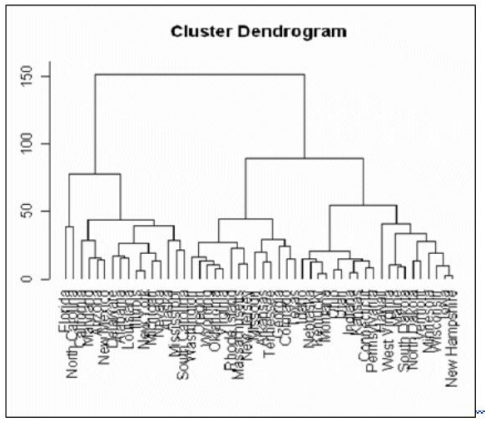
Divisive method

– Top-down approach – Views the entire dataset as a single cluster and partitions it – Continues partitioning until only one individual exists in every cluster • Examples:



Examples of the cluster method and phylogenetic trees





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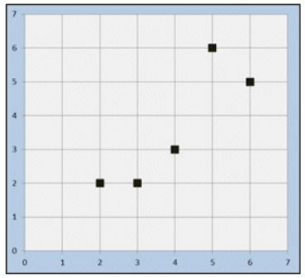
Connection-based hierarchical clustering analysis (1)

- Example: Center-based linkage, Clustering –

  Calculate distances for data

  – Merge 1 and 2 into cluster A, then calculate the centroid of cluster A –

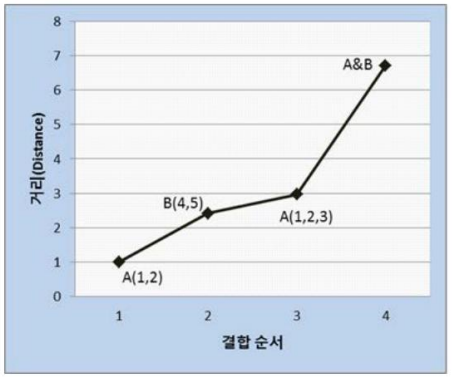
  Recalculate the distance matrix to find that 4 and 5 are the closest observations, forming cluster B – Repeat

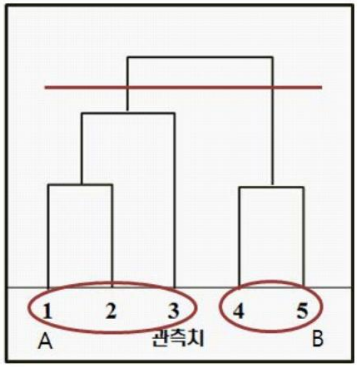


Obs.	X1	X2
1	2	2
2	3	2
3	4	3
4	6	5
5	6	6

거리 행렬

	1	2	3	4	5
1	0				
2	1	0			
3	2.2361	1.4142	0		
4	5	4.2426	2.8284	0	
5	5	4.4721	3.1623	1.4142	0



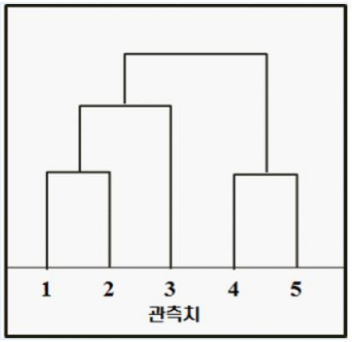


Obs.	X1	X2
A	2.5	2
3	4	3
4	6	5
5	5	6

	A	3	4	5
A	0			
3	1.8028	0		
4	4.6098	3.1623	0	
5	4.717	3.3166	1.4142	0

Obs.	X1	X2
A	2.5	2
3	4	3
B	5.5	5.5

	A	3	B
A	0		
3	1.8028	0	
B	4.6098	4.6098	0



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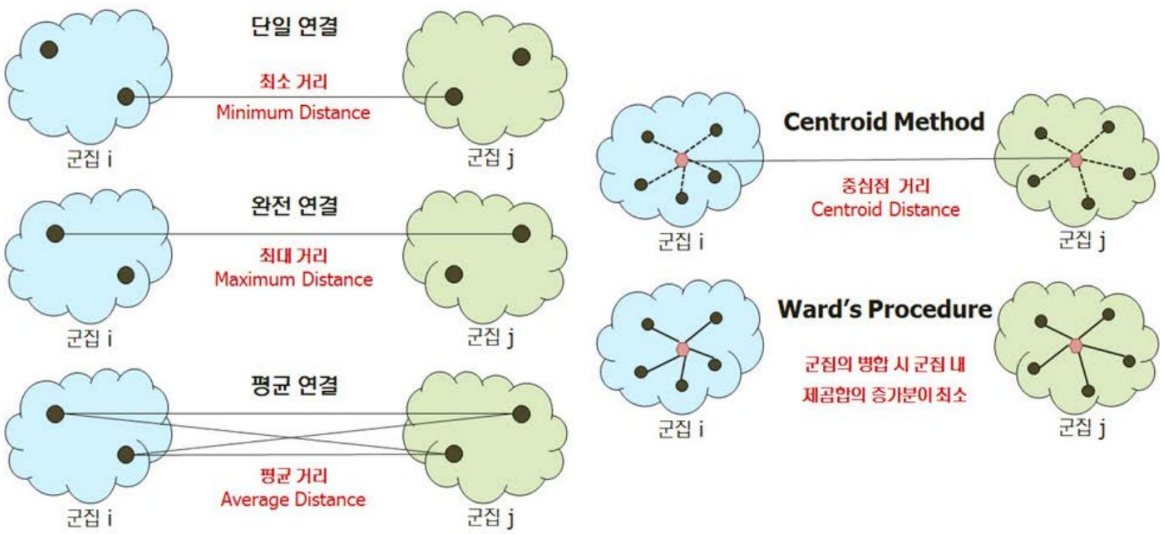
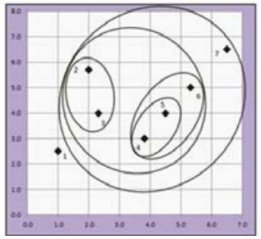
Connection-based hierarchical clustering analysis (2)

- Connection methods in hierarchical clustering analysis



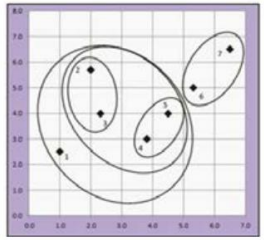
❖ Single Link or MIN :

- $dist(\{4,5\},\{2,3\}) = \min(d_{42}, d_{43}, d_{52}, d_{53})$
  $= \min(3.245, 1.803, 3.023, 2.2) = 1.803$



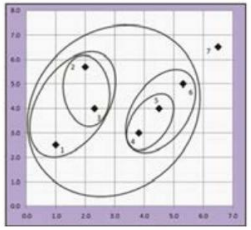
❖ Complete Link or MAX, CLIQUE :

- $dist(\{2,3\},\{4,5\}) = \max(d_{24}, d_{25}, d_{34}, d_{35})$
  $= \max(3.245, 3.023, 1.803, 2.2) = 3.245$



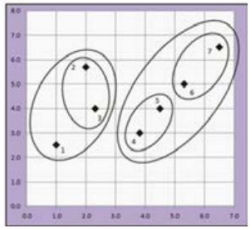
❖ Group Average :

- $dist(\{4,5\},\{6\}) = 1.89$
- $dist(\{4,5\},\{2,3\}) = 2.568$

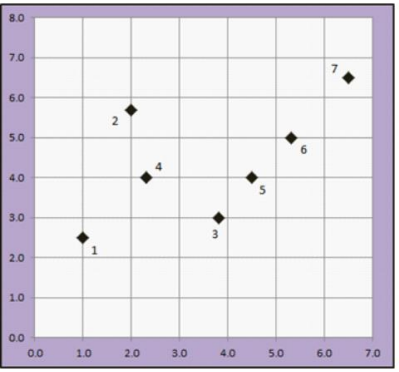


❖ Ward's Method :

- SSW 기반 연결

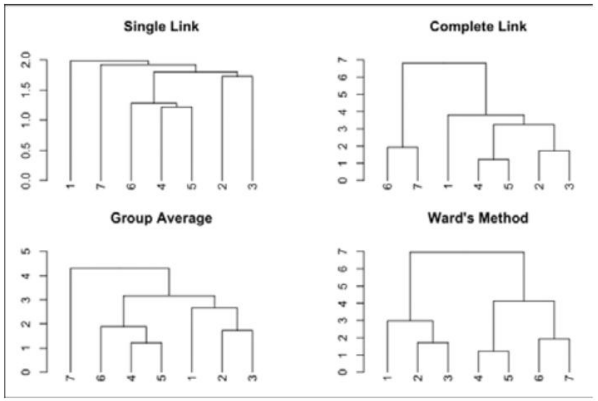


- example



Obs.	X1	X2
1	1.0	2.5
2	2.0	5.7
3	2.3	4.0
4	3.8	3.0
5	4.5	4.0
6	5.3	5.0
7	6.5	6.5

D(i,j)	1	2	3	4	5	6	7
1	0.000						
2	3.353	0.000					
3	1.985	1.726	0.000				
4	2.844	3.245	1.803	0.000			
5	3.808	3.023	2.200	1.221	0.000		
6	4.974	3.373	3.162	2.500	1.281	0.000	
7	6.801	4.571	4.888	4.420	3.202	1.921	0.000



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Center-based cluster analysis

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Basic Concepts of Center-Based Clustering

- Generally the most widely used cluster analysis method
- Assuming that there exists a data point representing each cluster
- Assign each observation to the data point with the closest representativeness

Unlike hierarchical clustering, it creates K clusters by partitioning once.

– After recalculating the centroids of the generated clusters, the observation reassignment process is repeated to obtain the final clusters

I go to find it.

- The center point is also called the centroid, prototype, or representative.
- Types of algorithms

– K-Means cluster analysis

- When using the average of the data as a representative data point

– K-Median Cluster Analysis

- When using the median

– K-Medoid Cluster Analysis (Reference)

- Since the mean, median, etc., are virtual points, use actual data points that are representative.

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K-Means Algorithm (1)

- K-Means Clustering Procedure: ÿ

  Determine the number of clusters

  K. ÿ Select the initial K cluster centroids. ÿ Assign each

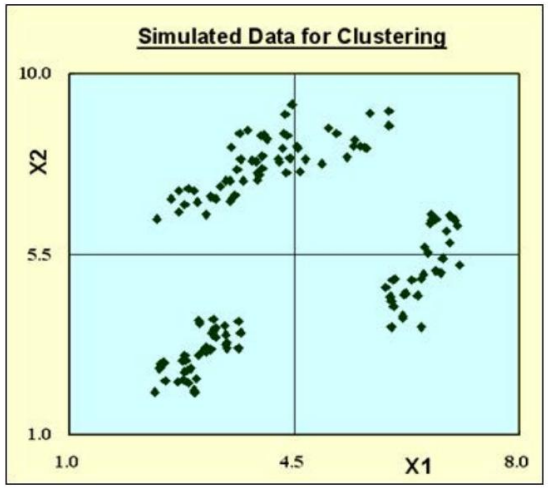
  observation to the cluster to which the nearest centroid belongs. ÿ Calculate

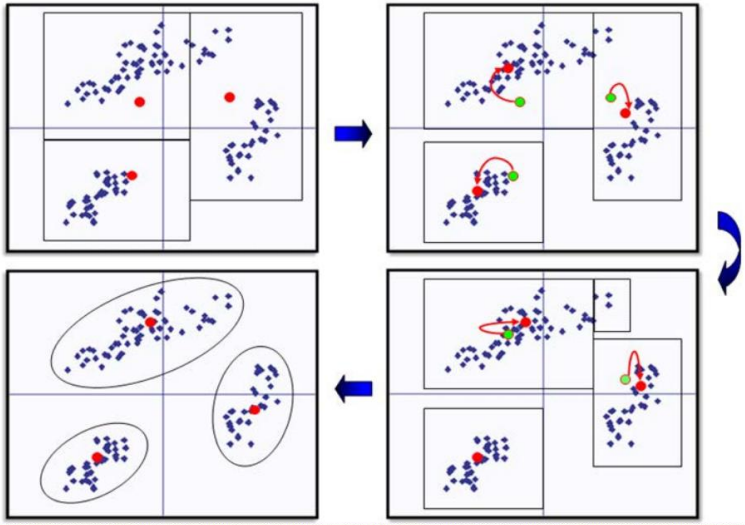
  new centroids for each cluster using the observations belonging to that cluster. ÿ

  Repeat steps ÿ through ÿ; if there is no difference between the existing centroids and the new

  centroids, the process is complete. – The user must provide the desired number of clusters K and K initial c

- Diagram of the K-Means algorithm procedure





The possibility that the result will vary

depending on the initial center point value

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K-Means Algorithm (2)

- Points to note when using the K-Means algorithm

  – The K-Means algorithm is a simple and easy-to-understand algorithm that finds local optima

  It is merely that, since cluster analysis fundamentally makes it nearly impossible to find the optimal cluster, there is no major problem.

  It might not be the case. Apply the K-Means algorithm multiple times while varying the number of clusters and initial centroids

  If you make a final selection after performing the analysis, it is an excellent cluster analysis method.

  – Since results may vary depending on the number of clusters and initial centroids, if outliers exist

  Caution should be exercised that strange clusters may be generated.

  – The problem of selecting the number of clusters K often relies on user experience or intuition, but prior statistics

  Methods involving analysis are also recommended. The first method involves performing hierarchical clustering to select the ***K*** value

  and initial centroids, while the second method involves using multivariate statistical analysis, such as principal component analysis, to

  reduce the data to two or three dimensions and then determining the ***K*** value using a graph .

  – Outliers distort the results of cluster analysis by the K-Means algorithm or SSE

  Since it increases the likelihood of outliers, it is recommended to perform cluster analysis after investigating and removing them beforehand.

  – Occurrence of empty clusters, that is, the K-Means algorithm has centroids to which no observations are assigned.

  You can do it.

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Cluster evaluation scale

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Categories of cluster analysis evaluation scales

ÿ External Validation Measures

– A measure that uses prior knowledge of the cluster or expert opinions not inherent in the data

Using uninformed information

– The difficulty of objective evaluation because there is no correct answer for the final solution

ÿ Internal Validation Measures

– Evaluation using only the data under analysis without referring to the cluster solution considered to be the correct answer

– Most cluster analysis situations fall under this category

– Utilizing the fact that within-cluster similarity should be large and between-cluster similarity should be small

– Within-cluster similarity is expressed as cohesion, and between-cluster similarity as separation.

ÿ Relative Validation Measures

– To find an appropriate number of clusters or to determine whether performing cluster analysis is meaningful

to use

– To find the appropriate number of clusters, the intra-cluster variance (SSW) or mean silhouette coefficient explained earlier

Use the back

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External evaluation metrics

- Purity
  - An indicator showing how much each evaluation cluster is composed of the same observations from the true cluster
  - Weighted average of the purity of each evaluation cluster
- Entropy
  - Measures used in the decision tree
- F-Measure
  - Harmonic mean of Precision and Recall
- etc
  - Number of True Positive (TP) pairs, Number of False Negative (FN) pairs, Number of Positive (FP) pairs, Number of True Negative (TN) pairs, etc.
  - Jaccard Coefficient calculated through TP, FN, FP, TN, etc., Land G

    Number (Rand Index, RI), Fowlkes-Mellows scale (FM Measure), etc.

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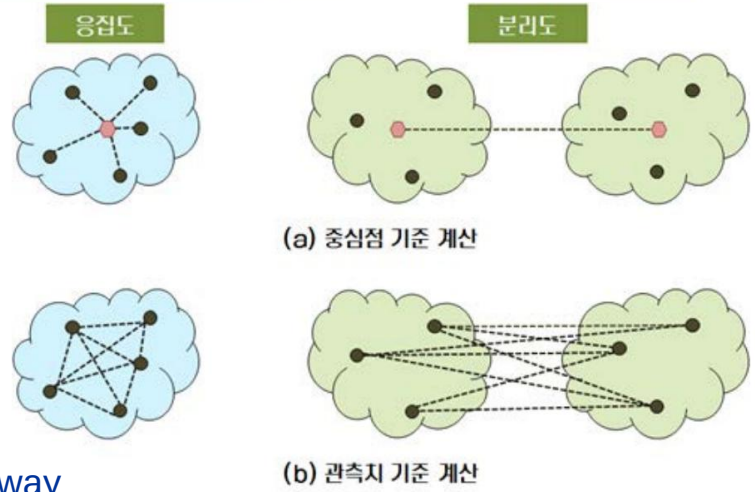
Internal evaluation scale

- Calculate the scale using the concepts of cohesion and separation
  - If cohesion is high,

    The cluster analysis results were positive.
  - If the separation is high,

    The cluster analysis results were negative



- A measure that combines cohesion and separation in an appropriate way
  - Dunn Indicator
  - CH indicators
  - DB Metrics
  - Brock indicator
  - Silhouette Coefficient: Calculate silhouette coefficients for each cluster, and the average silhouette coefficient between clusters

    Select the cluster with the largest number

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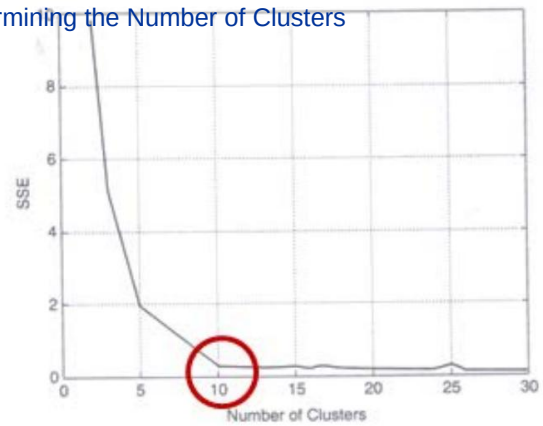
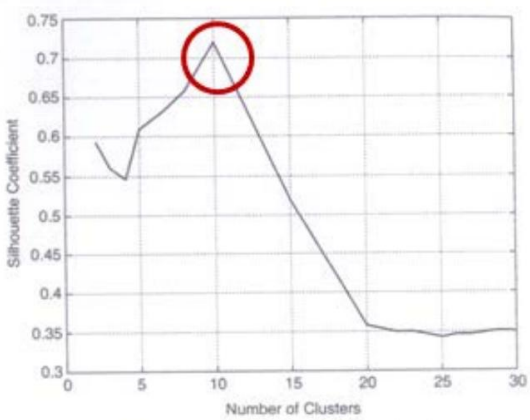
Relative evaluation scale

- Determine whether it is meaningful to find an appropriate number of clusters or perform cluster analysis used to do
- Within-cluster variance (SSW, the sum of the squared distances of each cluster) or mean silhouette coefficient is frequently used (other scales are also possible)
- Case Study: Determining the Number of Clusters



SSE versus number of clusters



Average silhouette coefficient versus number of clusters

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k-means clustering practice

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Data preparation (1)

- The data to be used for k-means clustering is from scikit-learn datasets

  Utilizing the provided iris data

  – The data to be used is data measuring the width and length of each part of the petals (150 records)

    - Number of records: 150, Number of fields: 4

    - Fields (Attributes): Sepal Length, Sepal Width,

      Petal Length, Petal Width

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Data preparation (2)

- Verification of data
  - Load Iris data from scikit-learn
  - Data in dictionary form
    - 150 items in order: Sepal Length, Sepal Width, Petal Length,

      Save Petal Width value
  - Check data in a Python program

```
from sklearn.datasets import load_iris
from sklearn.cluster import KMeans

#import numpy as np
import pandas as pd

iris = load_iris()

irisDF = pd.DataFrame(data=iris.data, columns=['sapal_length', 'sepal_width',
                      'petal_length', 'petal_width'])
irisDF.head(3)
```

	sapal_length	sepal_width	petal_length	petal_width
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2

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Clustering analysis

- Perform k-means clustering on the previously prepared data and verify the results (1) Cluster the iris data (150 cases) into 3 groups (0, 1, 2)
  - Results of clustering 150 data points into three groups: clusters 0, 1, and 2

```
kmeans = KMeans(n_clusters=3, init='k-means++', max_iter=300, random_state=0)
kmeans.fit(irisDF)
print(kmeans.labels_)
```

```
[1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 2 0 2 2 2 2 0 2 2 2 2
 2 2 0 0 2 2 2 2 0 2 0 2 0 2 2 0 0 2 2 2 2 2 0 2 2 2 2 0 2 2 2 0 2 2 2 0 2
 2 0]
```

(2) Cluster the iris data into 4 groups (0, 1, 2, 3)

```
kmeans = KMeans(n_clusters=4, init='k-means++', max_iter=300, random_state=0)
kmeans.fit(irisDF)
print(kmeans.labels_)
```

```
[1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
 1 1 1 1 1 1 1 1 1 1 1 1 1 1 3 3 3 0 3 0 3 0 3 0 0 0 0 3 0 3 0 0 3 0 3 0 3 3
 3 3 3 3 3 0 0 0 0 3 0 3 3 3 0 0 0 3 0 0 0 0 0 3 0 0 2 3 2 2 2 2 0 2 2 2 3
 3 2 3 3 2 2 2 2 3 2 3 2 3 2 2 3 3 2 2 2 2 2 3 3 2 2 2 3 2 2 2 3 2 2 2 3 3
 2 3]
```

(3) Cluster the iris data into 5 groups (0, 1, 2, 3, 4)

```
kmeans = KMeans(n_clusters=5, init='k-means++', max_iter=300, random_state=0)
kmeans.fit(irisDF)
print(kmeans.labels_)
```

```
[0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 0 0 0 0 0 0 4 4 4 2 4 2 4 2 4 2 2 2 2 4 2 4 2 2 4 2 4 2 4 4
 4 4 4 4 4 2 2 2 2 4 2 4 4 4 2 2 2 4 2 2 2 2 2 4 2 2 1 4 3 1 1 3 2 3 1 3 1
 1 1 4 1 1 1 3 3 4 1 4 3 4 1 3 4 4 1 3 3 3 1 4 4 3 1 1 4 1 1 1 4 1 1 1 4 1
 1 4]
```